Computer exercise 2

1 Preparations

Study the theory related to the KKT conditions, as well as the theory of penalty methods, in the textbook.

Please also note that you need to have the MATLAB Optimization Toolbox installed; for instructions, see `https://se.mathworks.com/products/optimization.html`.

2 Constrained optimization: KKT conditions & penalty methods

Consider the problem to

$$\begin{aligned} \text{minimize} \quad & f(x), \\ \text{subject to} \quad & g_i(x) \leq 0, \quad i = 1, \ldots, m, \end{aligned}$$

where f and g_i, for $i = 1, \ldots, m$, are continuously differentiable functions. Penalty methods are generally of one of two different kinds: either *exterior* or *interior* penalty methods, depending on if the methods generally give an infeasible of strictly feasible sequence of iteration points. We have implemented one method of each kind in MATLAB.

The *exterior* penalty method (sometimes called just the "penalty method") works with the relaxation

$$\underset{x \in \Re^n}{\text{minimize}} \; f(x) + \rho_k \psi(x),$$

where $\rho_k > 0$ and $\rho_k \to +\infty$ when $k \to +\infty$, and the penalty function is the quadratic function

$$\psi(x) := \sum_{i=1}^{m} \left(\max\{0, g_i(x)\}\right)^2.$$

The *interior* penalty method (sometimes called the "barrier method") works with the relaxation

$$\underset{x \in \Re^n}{\text{minimize}} \; f(x) + \mu_k \phi(x),$$

where $\mu_k > 0$ and $\mu_k \to 0$ when $k \to +\infty$, and where the penalty function is the function

$$\phi(x) := -\sum_{i=1}^{m} \log(-g_i(x)).$$

Important: In order to avoid numerical problems one usually lets the sequences ρ_k and μ_k converge *slowly*. In this computer exercise, you will be the one varying the parameters; *you should resepct this rule*. Changing the parameters drastically between iterations might lead to behaviour that is not representative for how the methods work in practise.

Description of the interface

To start, download the zip-file from the course homepage and follow the directions given. Move to the directory `Files_computer_exercise_2` and start MATLAB by simply typing `matlab`. In order to run the programs, you should move from the directory `Files_computer_exercise_2` to one of the subdirectories `kkt`, `epa` (exterior penalty algorithm), or `lipa` (interior penalty, or interior point, algorithm for linear programming). In any of these directories, start the corresponding exercise by typing `go` in Matlab's command window.

When running the program in the subdirectory `kkt`: after choosing the example from the drop-down list in the upper-right part of the window, press the "Load" button. You will see a plot with level curves of the cost function (the color gradient in the background; yellow is higher value, green is lower), green arrows that show the negative gradient of the cost function, and white lines that depicts the constraints of the problem. When clicking on the plot, a large green arrow will be shown; this is the negative gradient of the cost function in the point. If you click on an infeasible point, the program will tell you so under the "Load" button. If you click on one or more constraints, a red arrows will be shown; these are the normal vectors of the constraints clicked.

Note! there is, by necessity, a built-in tolerance in the graphical interface; if a constraint does not activate when you click on it, click again close by or try to give the coordinates to the point.

When running the programs in the the subdirectories `epa` or `lipa`: after choosing the example from the drop-down list in the lower-left part of the window, press the "Load" button. In the left window you will see the level sets of the objective function (you can think of them as a topographic map), the directions the *negative* gradient, as well as the curves constraining the feasible set. The coordinates of the current iteration point can be read below the left window.

After adjusting the desired penalty value, press the "Optimize" button. The right window will show the level sets of the penalised function, exactly as the algorithm would "see" it, were it not near-sighted. The current iteration point is plotted in the right window (pink "x" cross); the left window will contain the optimization "path", showing the progress of the algorithm (pink curve).

Note! In our implementation of EPA we solve the penalised problem using a gradient algorithm to obtain a locally optimal solution. Instead, one can perform only a few iterations of the gradient algorithm.

In IPA, we perform *only one* iteration of the modified Newton method (with an Armijo line search). We show the global minimum point in the right window using a red "o" circle; its evolution as the penalty parameter changes (so-called "central path") is shown in the left window as a red line.

Hint! You can change the starting point for the algorithm by modifying the variables `x1_start` and `x2_start` in the `.m`-file, corresponding to the example you solve. Even more, you can add your own problems by providing a corresponding `example*.m` file!

Exercises

There are four nonlinear problems, two convex (`example_nl{01,02}.m`), and two non-convex (`example_nl{03,04}.m`), as well as three linear problems (`example_lin[01-03].m`); you can find the problem formulations in the Appendix. *You will need these analytic formulations for some of the analysis that you have to do!*

1. Using the interior point method, solve the LPs 01–03. Do we always find an optimal extreme point (*Hint!* problem 03)? Notice how the algorithm follows closely the central path and goes "directly" to the global minimum point (i.e., it skips visiting the extreme points), if you change the penalty parameter smoothly. Compare this with how the Simplex method works.

2. Change the directory to `Files_computer_exercise_2/kkt` and type `go` at the Matlab prompt. Find the KKT points of the nonlinear problems (`example_nl[01-04].m`). Are the KKT conditions sufficient for the global optimality? (*Hint!* problem 03). Are they necessary? (*Hint!* problem 04).

   Hint: There is a built-in tolerance in the graphical interface that sometimes fools you. Make sure to analytically verify the results. This hint is especially important for the problem 04.

3. Using the exterior penalty algorithm, solve the nonlinear and linear problems. Can you get different "optimal" solutions by changing the penalty parameter in a different manner or by starting from different points? (*Hint!* nonlinear problem 03). Can you think of a reason for the slow convergence in nonlinear problem 04? (*Hint!* KKT)

Appendix

Linear problems

`example_lin01.m`

$$\begin{aligned} &\min\ x_1 + 3x_2, \\ &\text{s.t.} \begin{cases} x_1 + 2x_2 \geq 2, \\ x_1 - 3x_2 \leq 2, \\ -x_1 + 3x_2 \leq 12, \\ x_1, x_2 \geq 0, \end{cases} \end{aligned}$$

`example_lin02.m`

$$
\begin{aligned}
&\min\ x_1 + 3x_2, \\
&\text{s.t.}\ \begin{cases}
1x_1 + \frac{1}{10}x_2 \geq 1, \\
\frac{1}{2}x_1 + \frac{1}{9}x_2 \geq 1, \\
\frac{1}{3}x_1 + \frac{1}{8}x_2 \geq 1, \\
\frac{1}{4}x_1 + \frac{1}{7}x_2 \geq 1, \\
\frac{1}{5}x_1 + \frac{1}{6}x_2 \geq 1, \\
\frac{1}{6}x_1 + \frac{1}{5}x_2 \geq 1, \\
\frac{1}{7}x_1 + \frac{1}{4}x_2 \geq 1, \\
\frac{1}{8}x_1 + \frac{1}{3}x_2 \geq 1, \\
\frac{1}{9}x_1 + \frac{1}{2}x_2 \geq 1, \\
\frac{1}{10}x_1 + 1x_2 \geq 1, \\
x_1 \leq 20, \\
x_2 \geq 0.
\end{cases}
\end{aligned}
$$

`example_lin03.m`

$$
\begin{aligned}
&\min\ x_2, \\
&\text{s.t.}\ \begin{cases}
1x_1 + \frac{1}{10}x_2 \geq 1, \\
\frac{1}{2}x_1 + \frac{1}{9}x_2 \geq 1, \\
\frac{1}{3}x_1 + \frac{1}{8}x_2 \geq 1, \\
\frac{1}{4}x_1 + \frac{1}{7}x_2 \geq 1, \\
\frac{1}{5}x_1 + \frac{1}{6}x_2 \geq 1, \\
\frac{1}{6}x_1 + \frac{1}{5}x_2 \geq 1, \\
\frac{1}{7}x_1 + \frac{1}{4}x_2 \geq 1, \\
\frac{1}{8}x_1 + \frac{1}{3}x_2 \geq 1, \\
\frac{1}{9}x_1 + \frac{1}{2}x_2 \geq 1, \\
\frac{1}{10}x_1 + 1x_2 \geq 1, \\
x_1 \leq 20, \\
x_2 \geq 0.
\end{cases}
\end{aligned}
$$

Nonlinear problems

`example_nl01.m`

$$\begin{aligned} &\min\ x_1^2 + x_2^2, \\ &\text{s.t.} \begin{cases} x_1 \geq 2, \\ x_2 \geq 1, \\ \frac{1}{2}x_1 + \frac{1}{4}x_2 \leq 2. \end{cases} \end{aligned}$$

`example_nl02.m`

$$\begin{aligned} &\min\ x_1^2, \\ &\text{s.t.} \begin{cases} x_1 \geq 2, \\ x_2 \geq 1, \\ \frac{1}{2}x_1 + \frac{1}{4}x_2 \leq 2. \end{cases} \end{aligned}$$

`example_nl03.m`

$$\begin{aligned} &\min\ x_1 \sin(x_1) + x_2 \sin(x_2), \\ &\text{s.t.} \begin{cases} x_1 \geq \frac{1}{3}, \\ x_2 \geq \frac{3}{4}, \\ x_1 - \sin(x_2) \geq 0, \\ (x_1 - 1)^2 + (x_2 - 1)^2 \leq 5. \end{cases} \end{aligned}$$

`example_nl04.m`

$$\begin{aligned} &\min\ (x_1 + 1)^2 + \tfrac{1}{2}x_2^2, \\ &\text{s.t.} \begin{cases} x_1 \leq 3, \\ x_2 \geq 0, \\ \frac{1}{8}x_1^3 - x_2 \geq 0. \end{cases} \end{aligned}$$